

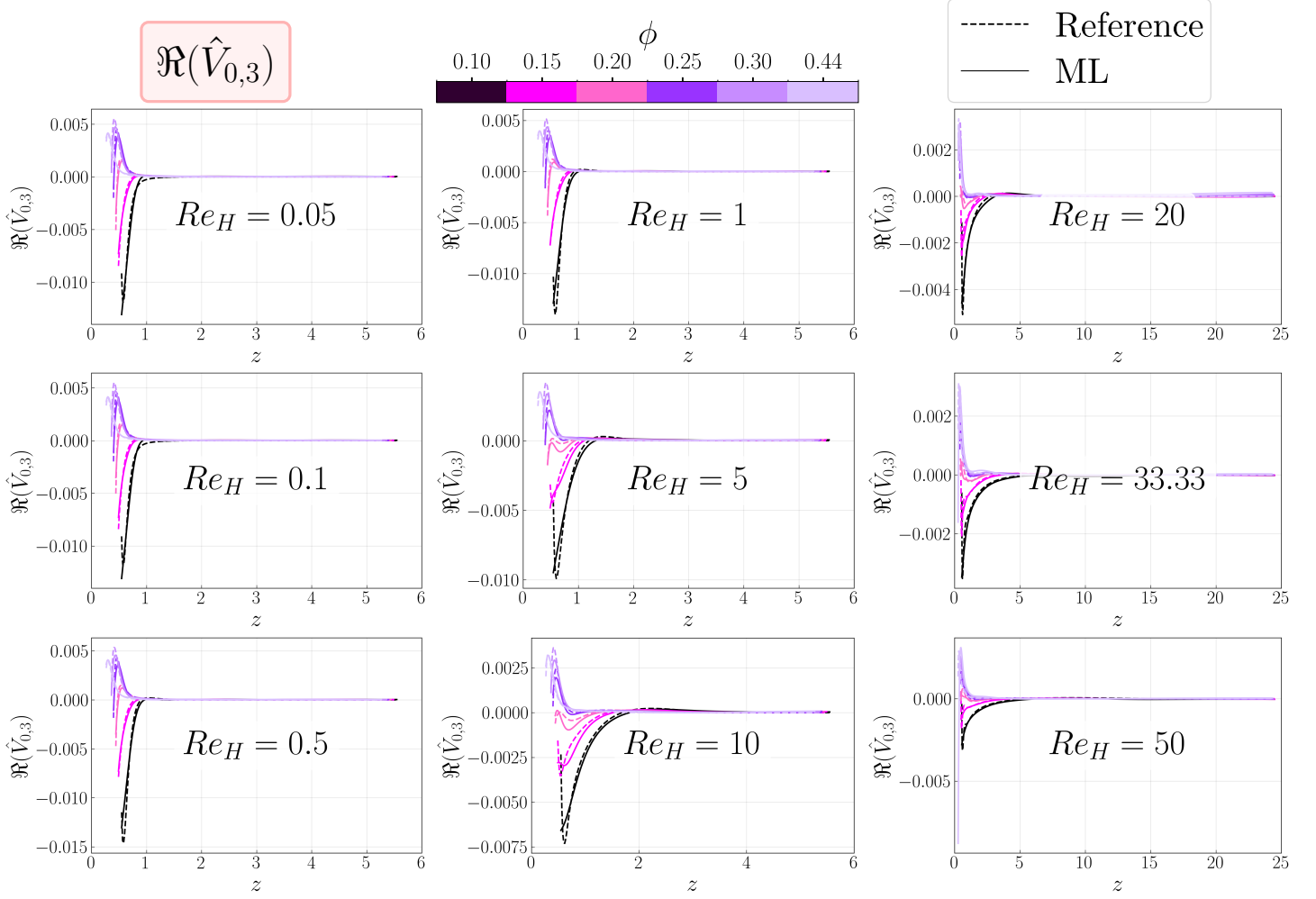


Figure C.24: Comparison of the reference DFT coefficient $\Re(\hat{V}_{0,3})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{0,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

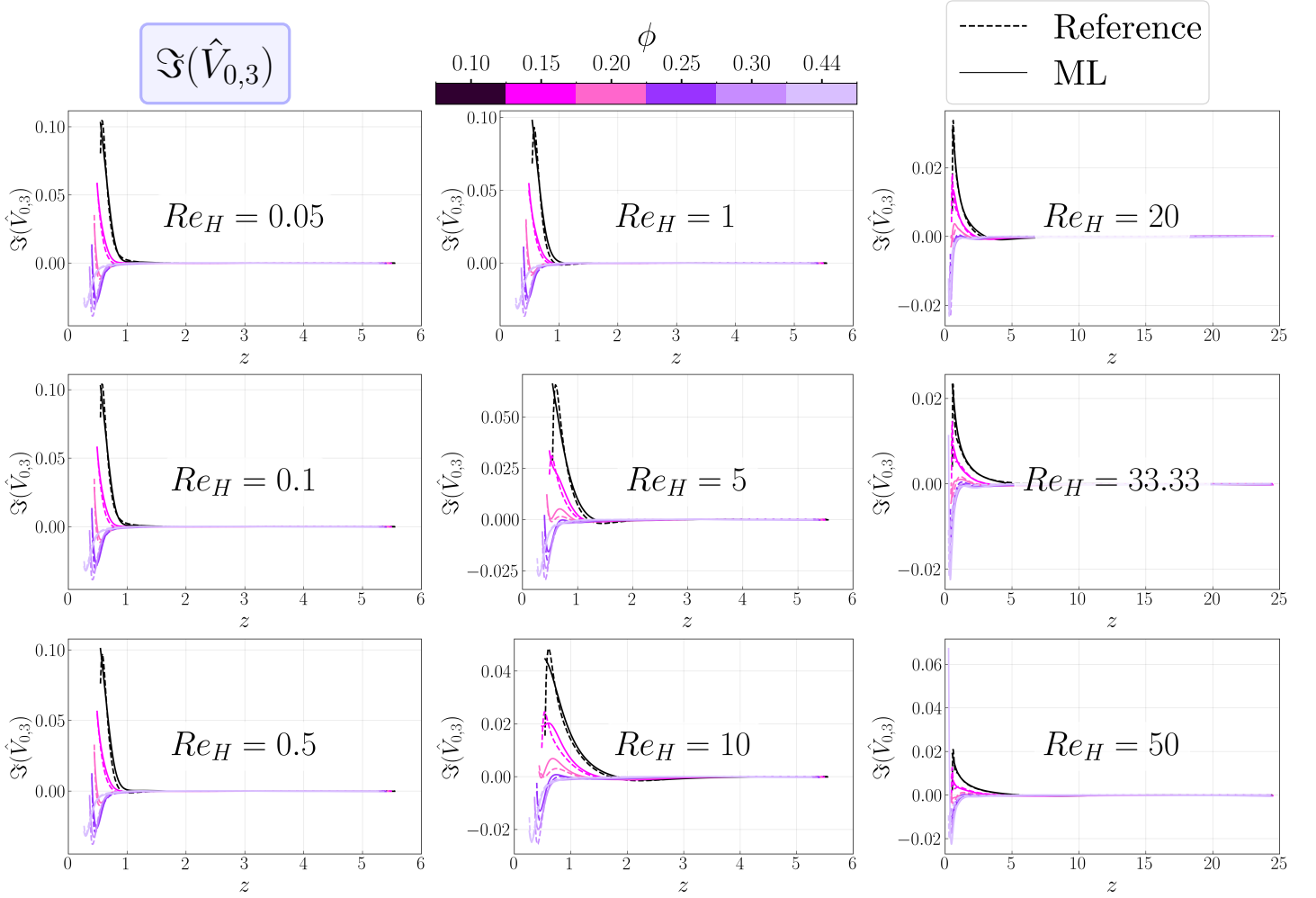


Figure C.25: Comparison of the reference DFT coefficient $\Im(\hat{V}_{0,3})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{0,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

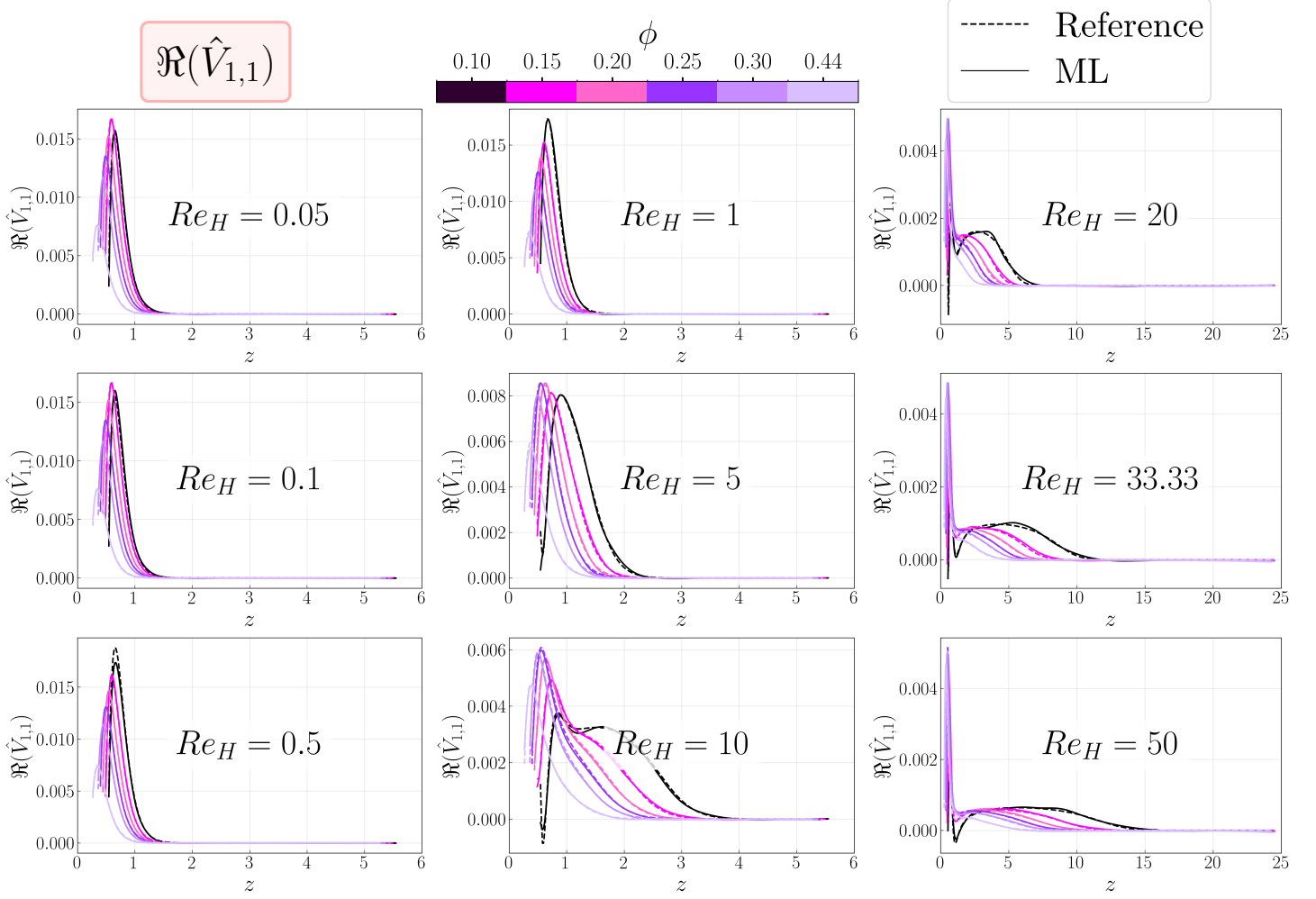


Figure C.26: Comparison of the reference DFT coefficient $\Re(\hat{V}_{1,1})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{1,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

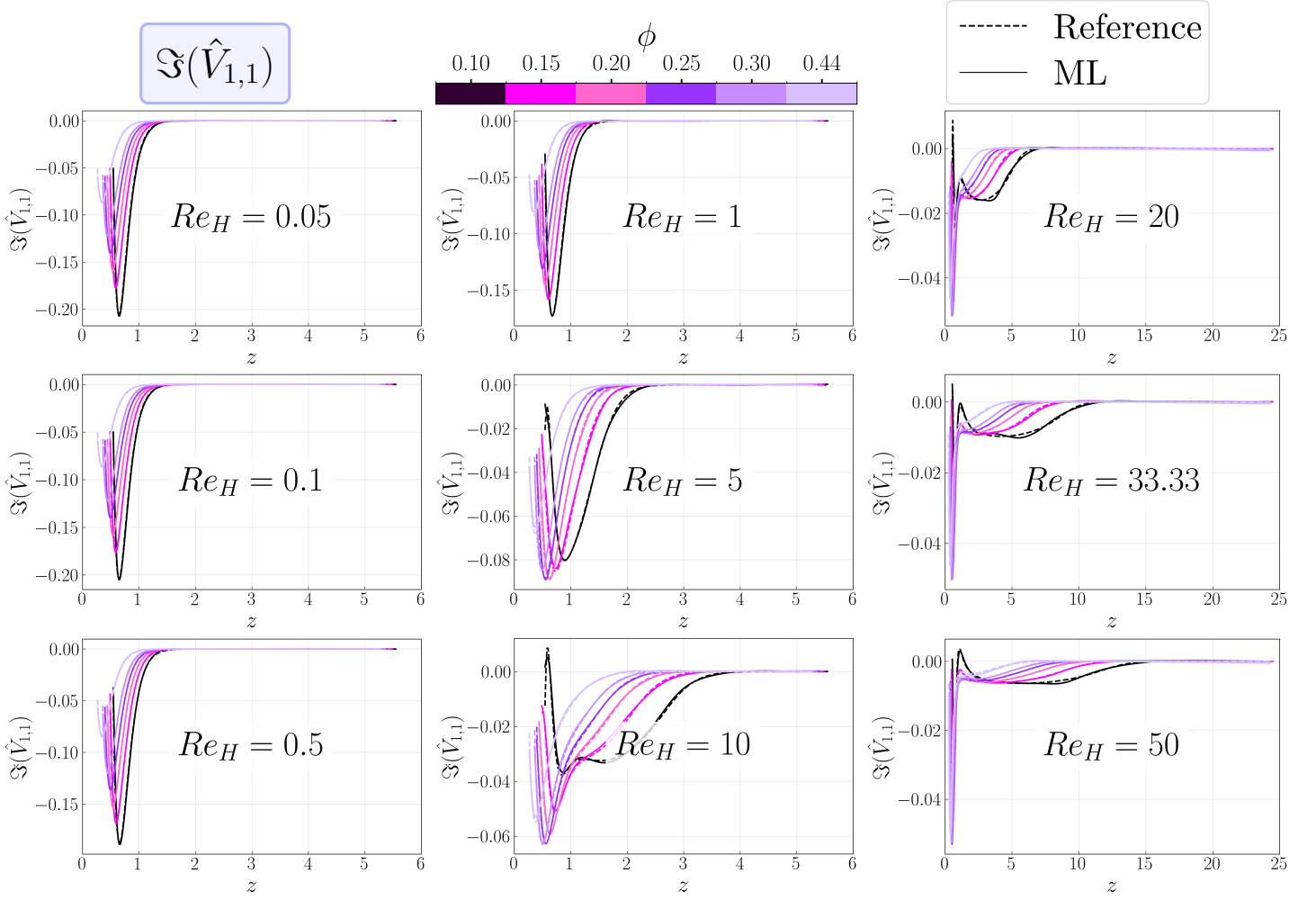


Figure C.27: Comparison of the reference DFT coefficient $\Im(\hat{V}_{1,1})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{1,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

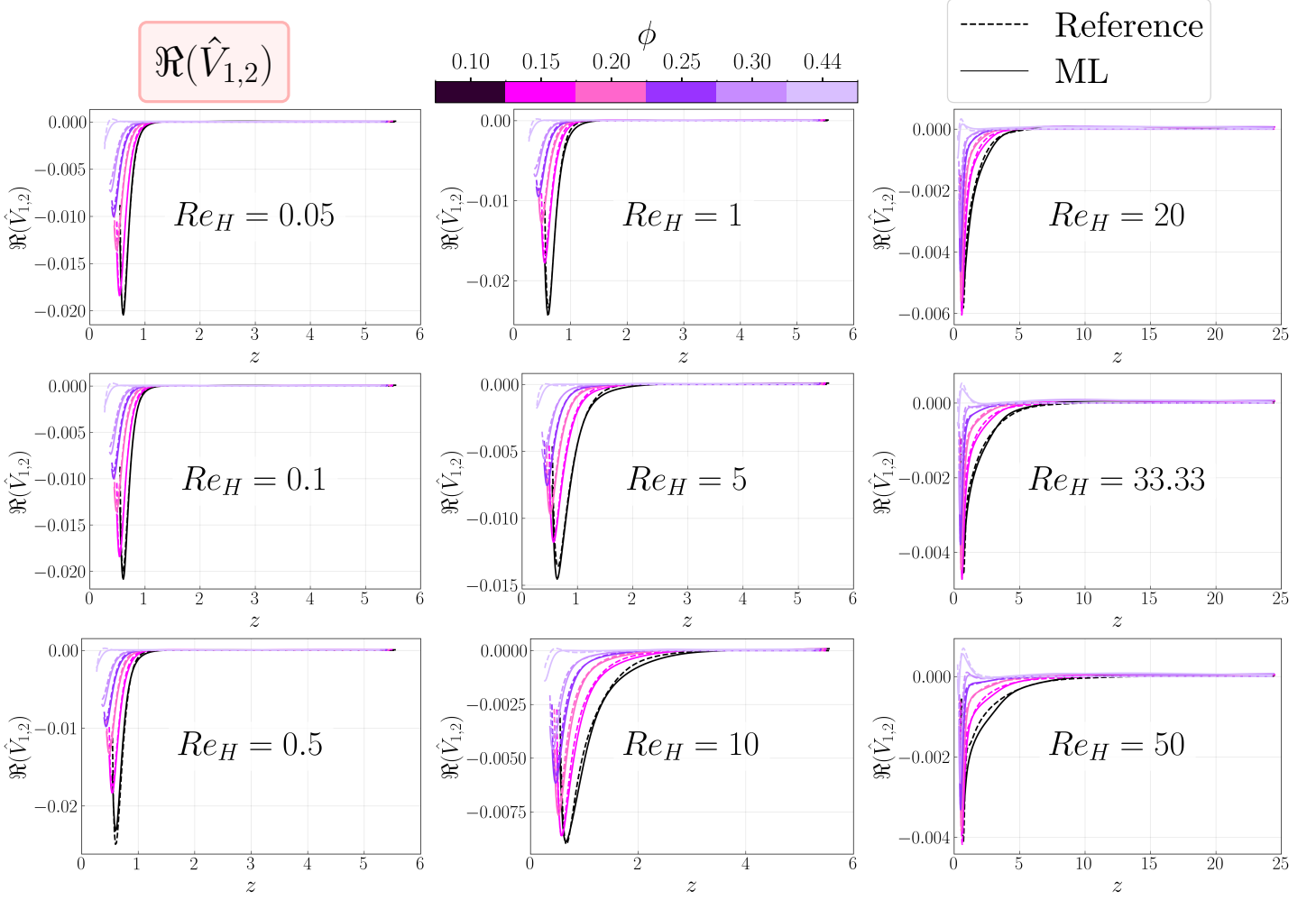


Figure C.28: Comparison of the reference DFT coefficient $\Re(\hat{V}_{1,2})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{1,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

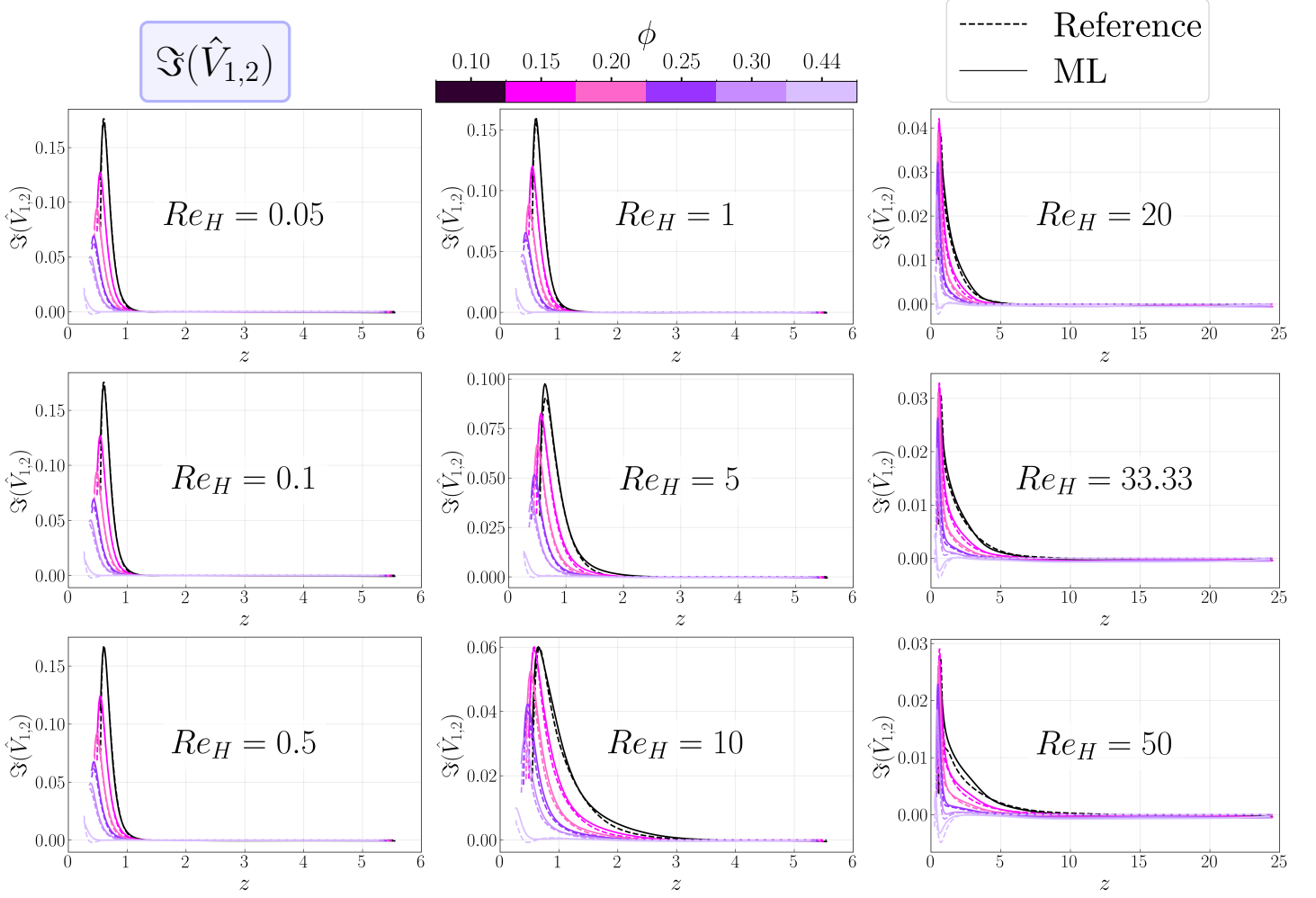


Figure C.29: Comparison of the reference DFT coefficient $\Im(\hat{V}_{1,2})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{1,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

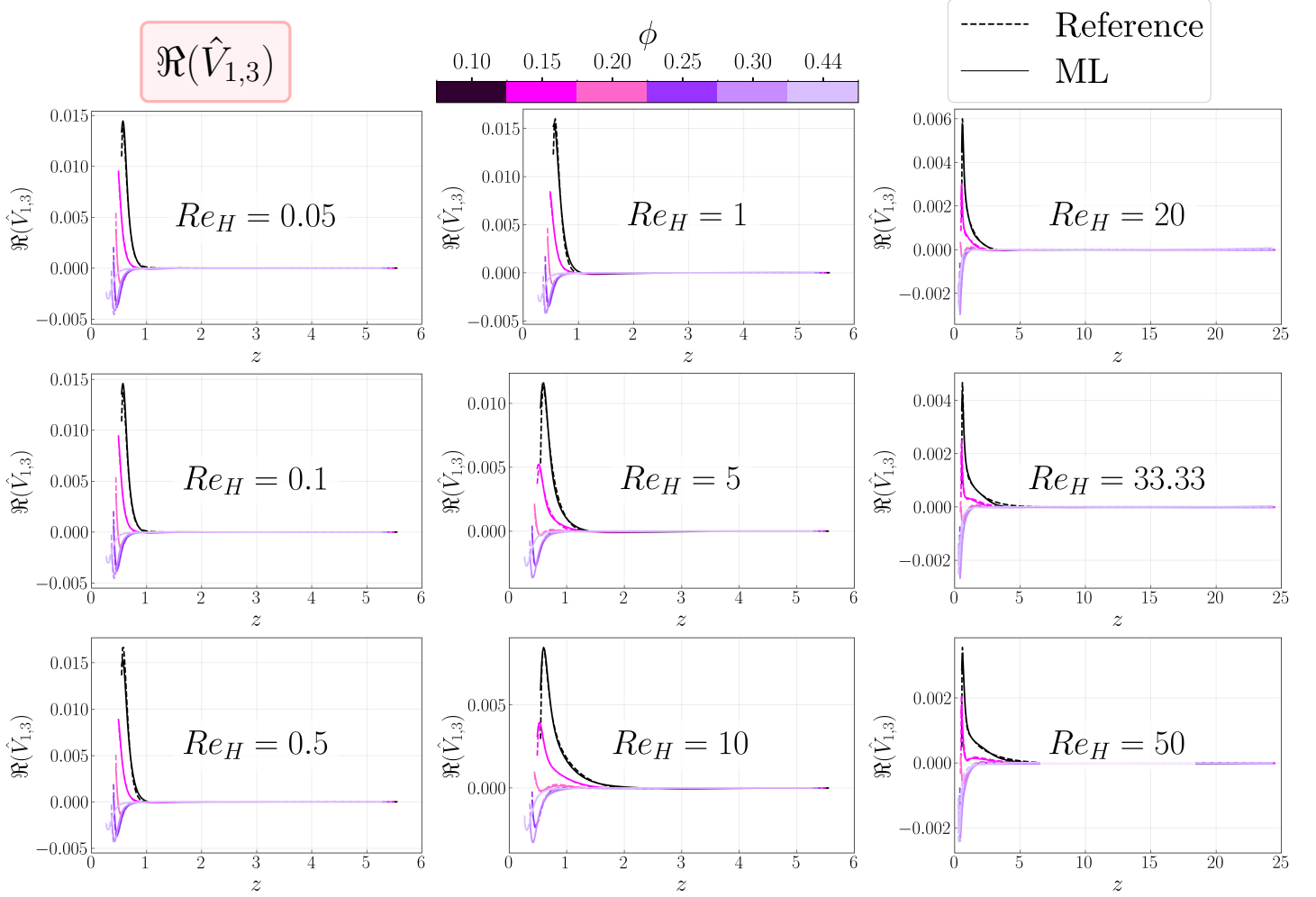


Figure C.30: Comparison of the reference DFT coefficient $\Re(\hat{V}_{1,3})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{1,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

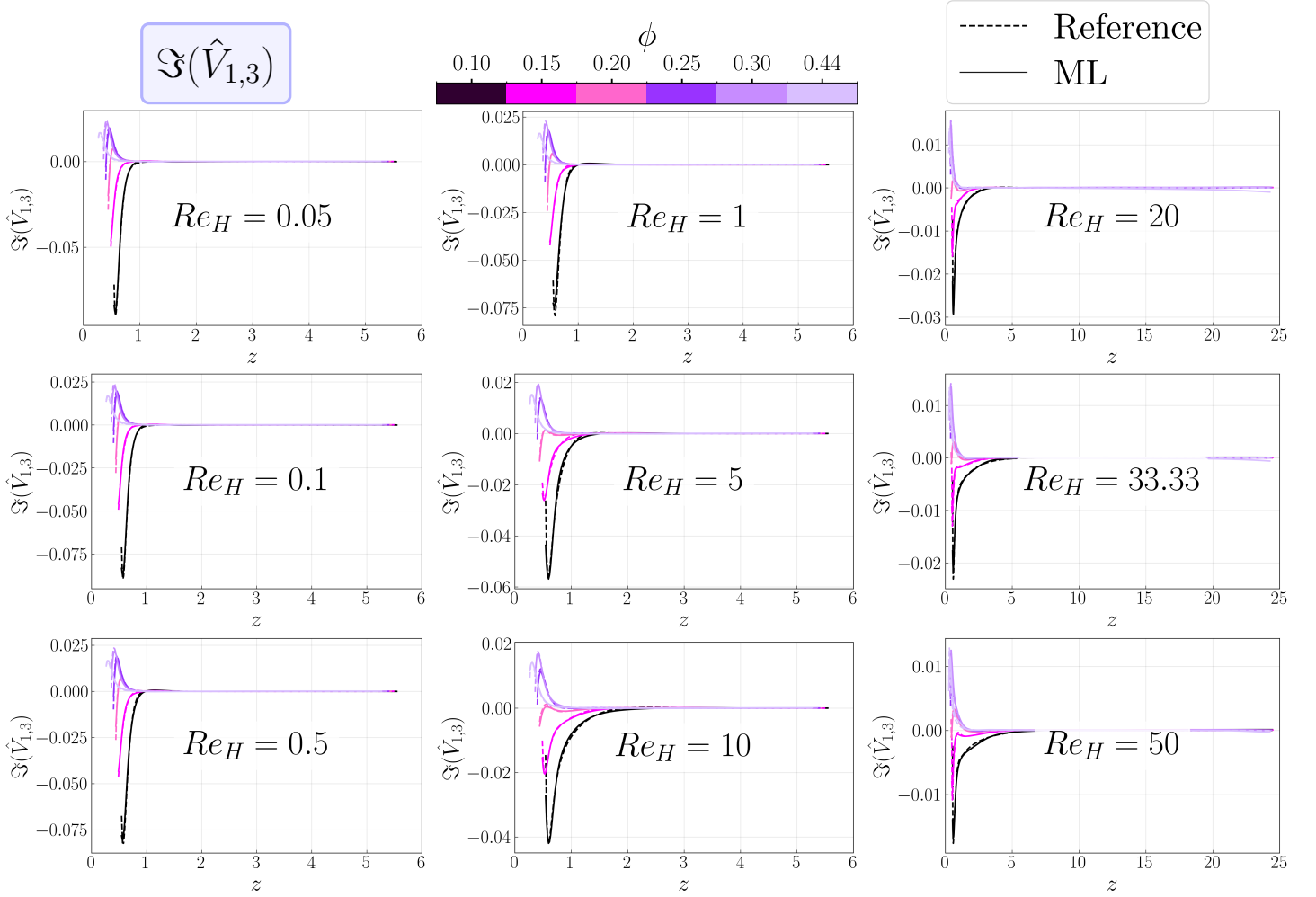


Figure C.31: Comparison of the reference DFT coefficient $\Im(\hat{V}_{1,3})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{1,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

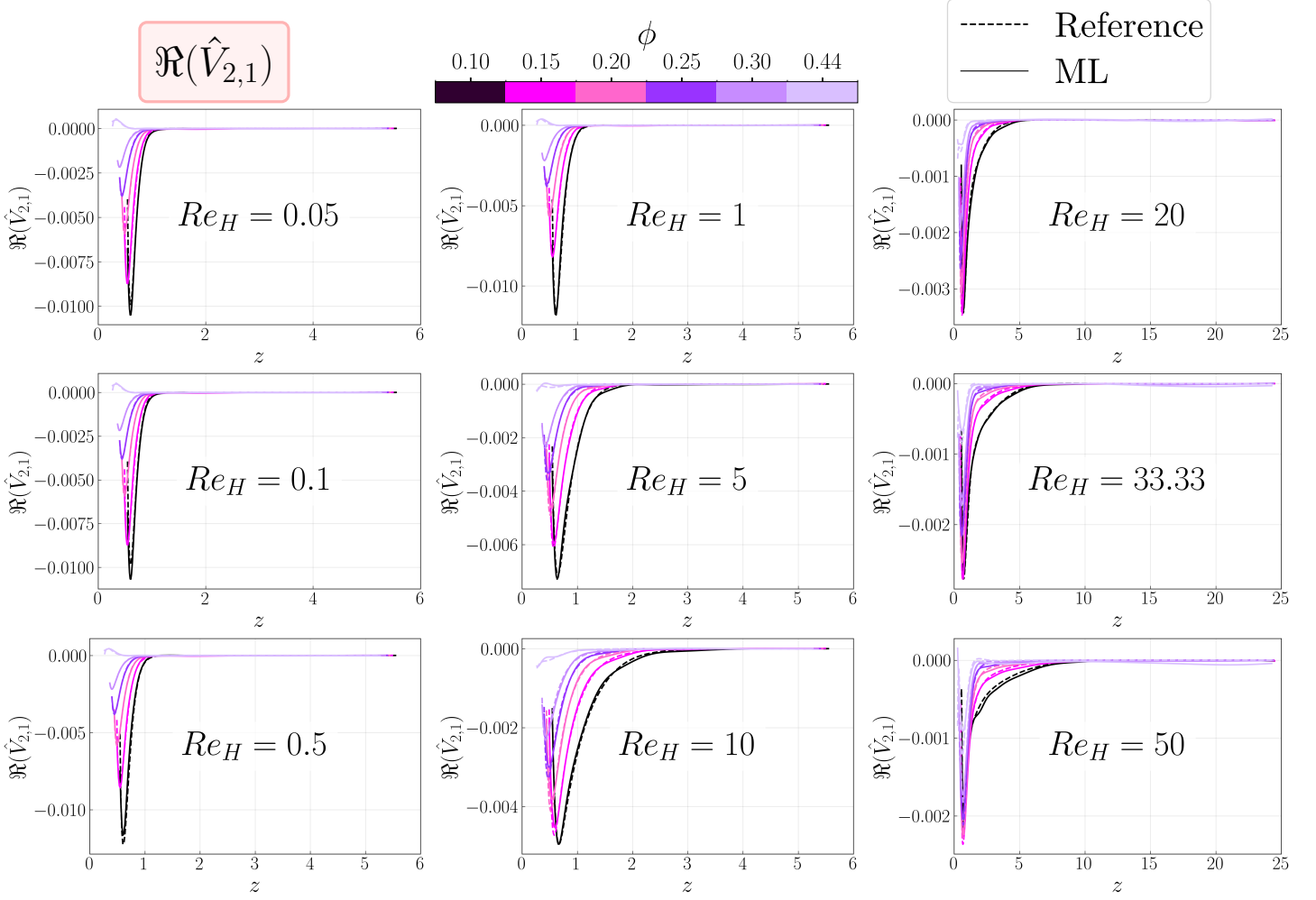


Figure C.32: Comparison of the reference DFT coefficient $\Re(\hat{V}_{2,1})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{2,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

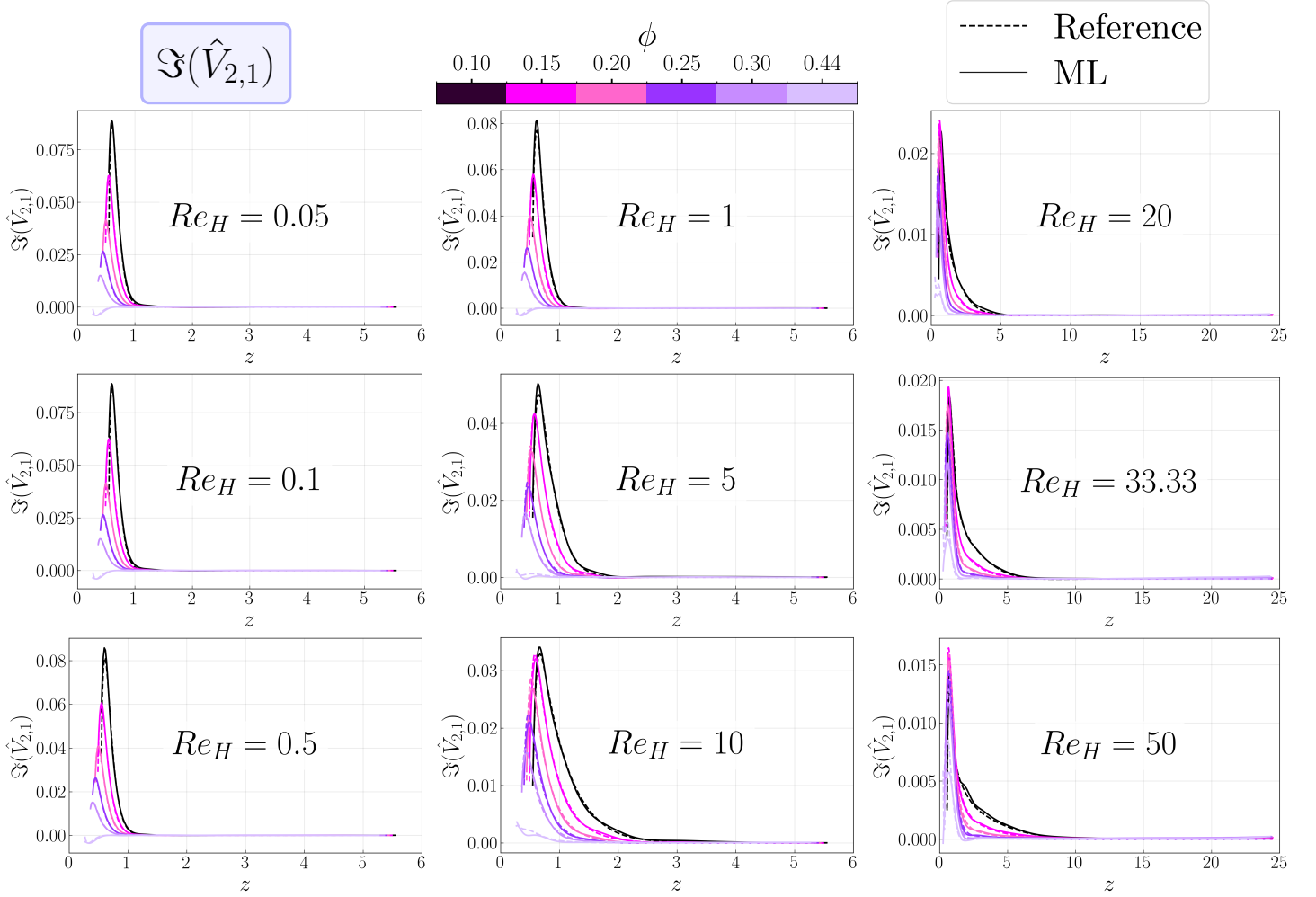


Figure C.33: Comparison of the reference DFT coefficient $\Im(\hat{V}_{2,1})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{2,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

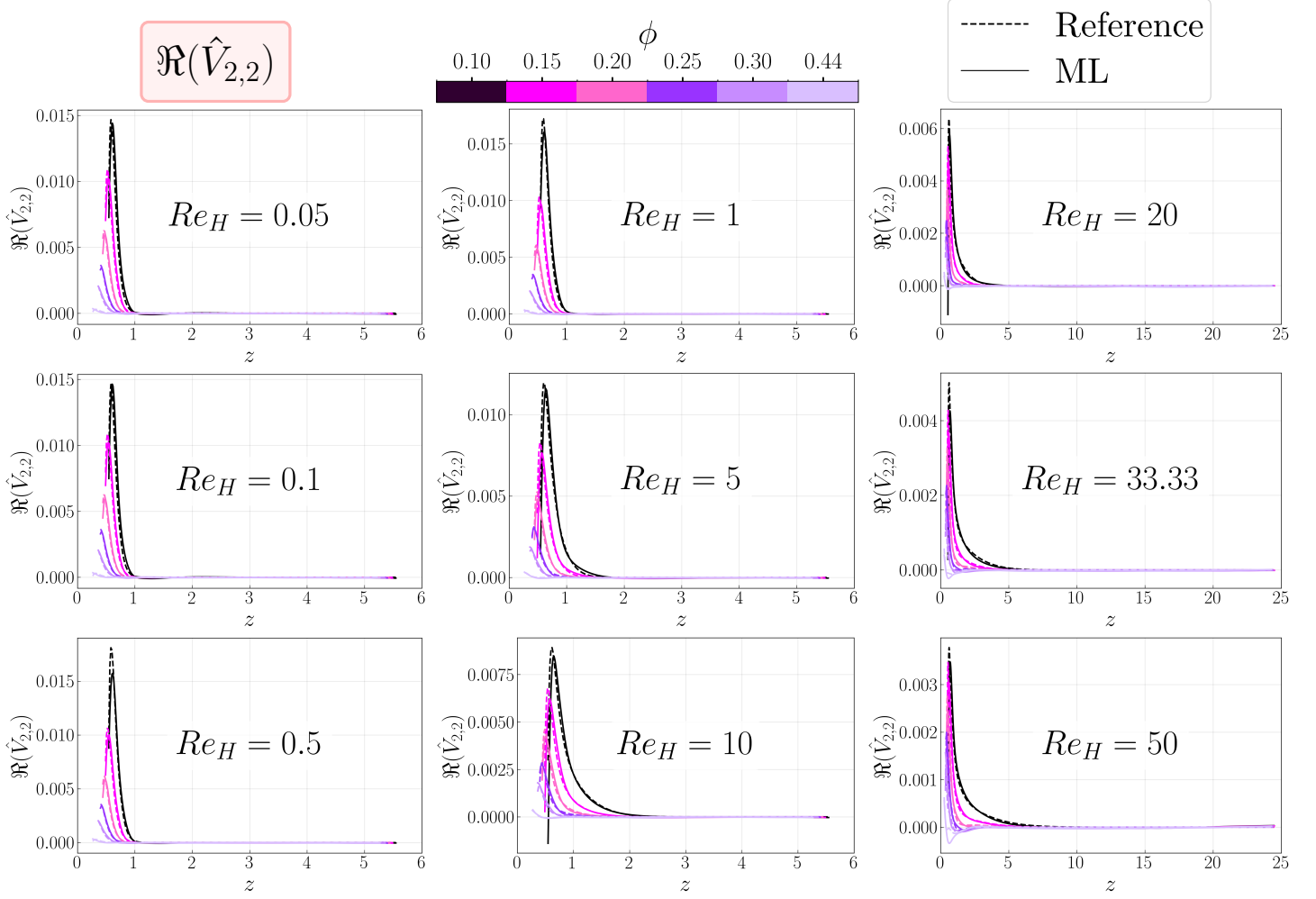


Figure C.34: Comparison of the reference DFT coefficient $\Re(\hat{V}_{2,2})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{2,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

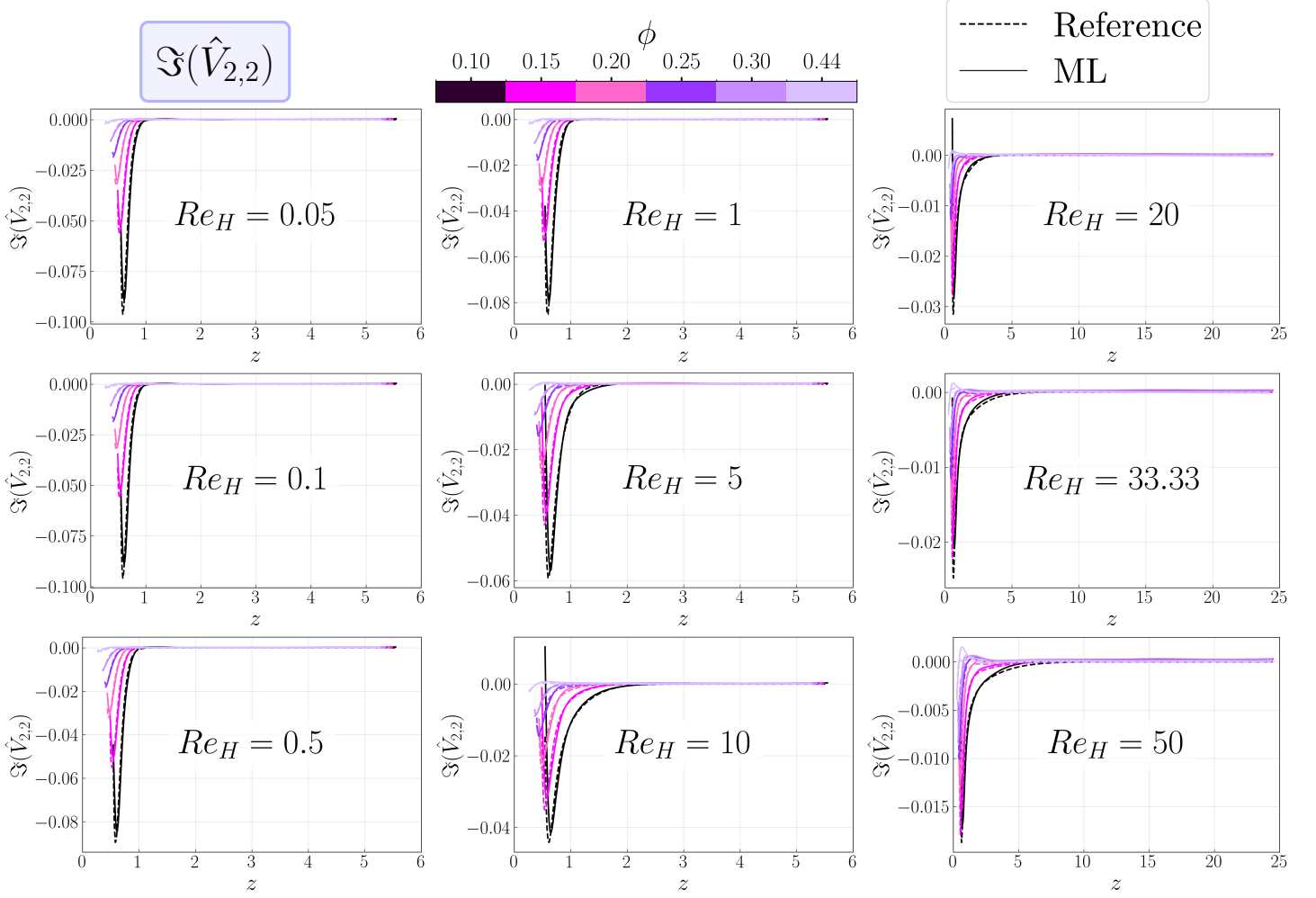


Figure C.35: Comparison of the reference DFT coefficient $\Im(\hat{V}_{2,2})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{2,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

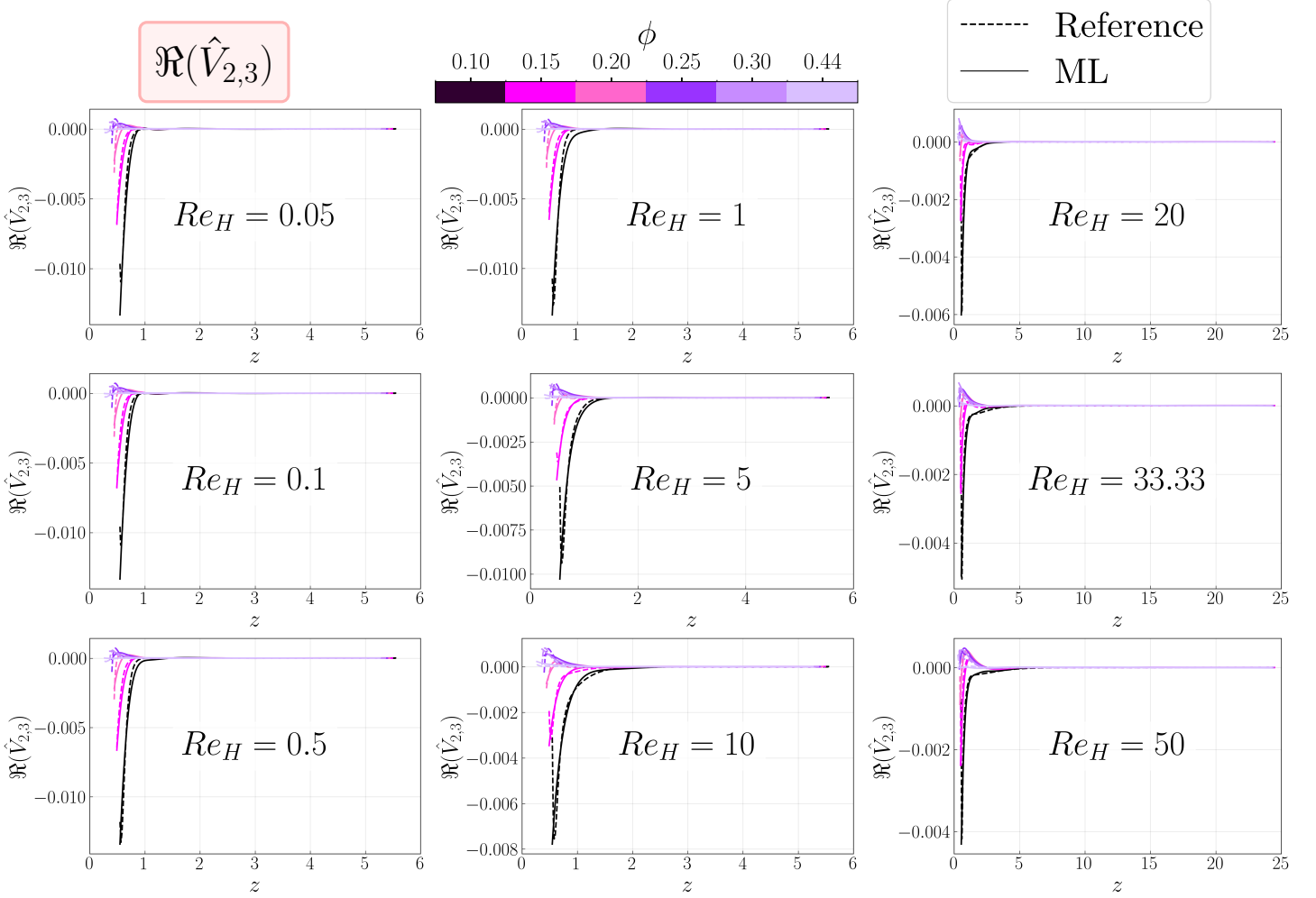


Figure C.36: Comparison of the reference DFT coefficient $\Re(\hat{V}_{2,3})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{2,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

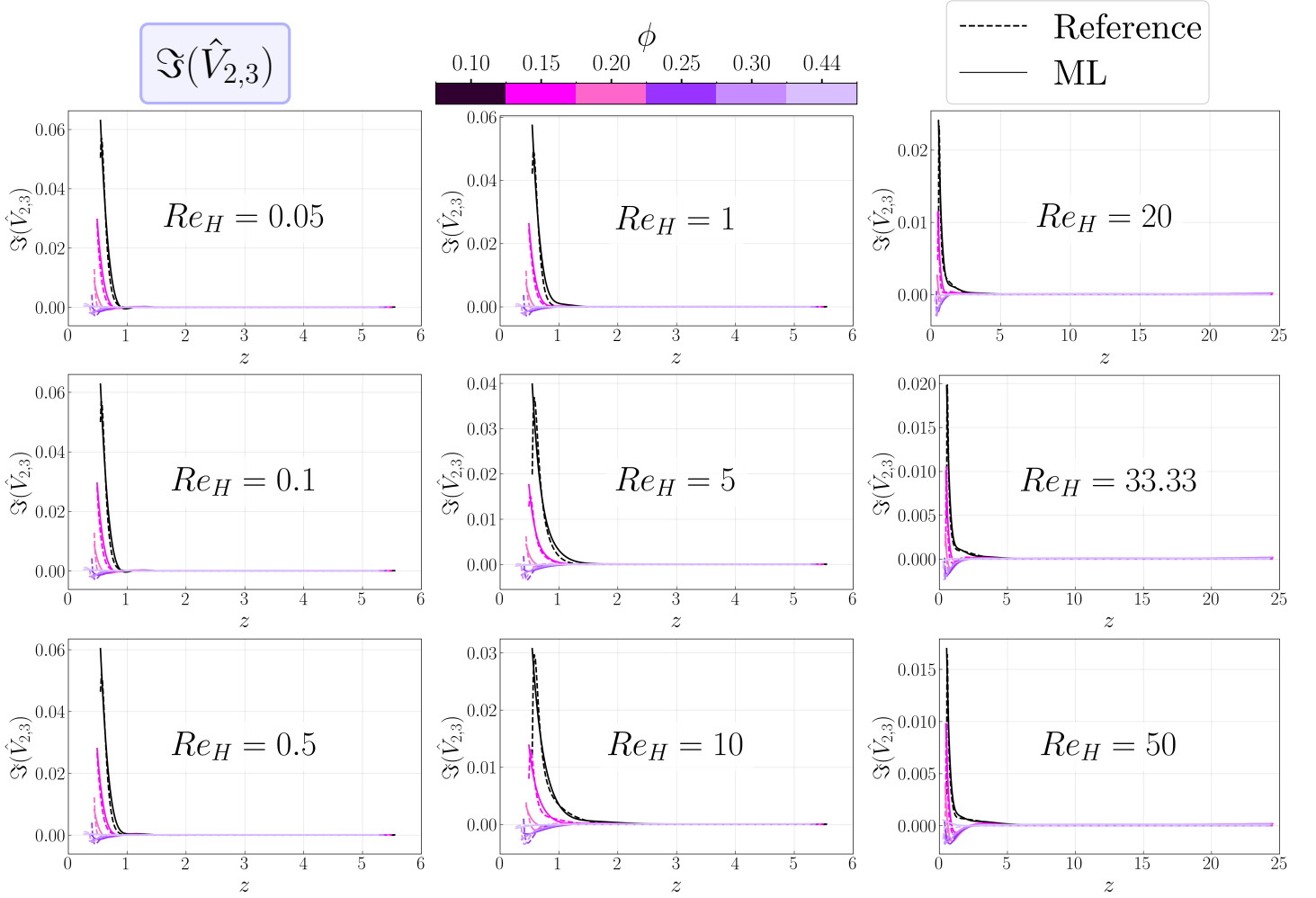


Figure C.37: Comparison of the reference DFT coefficient $\Im(\hat{V}_{2,3})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{2,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

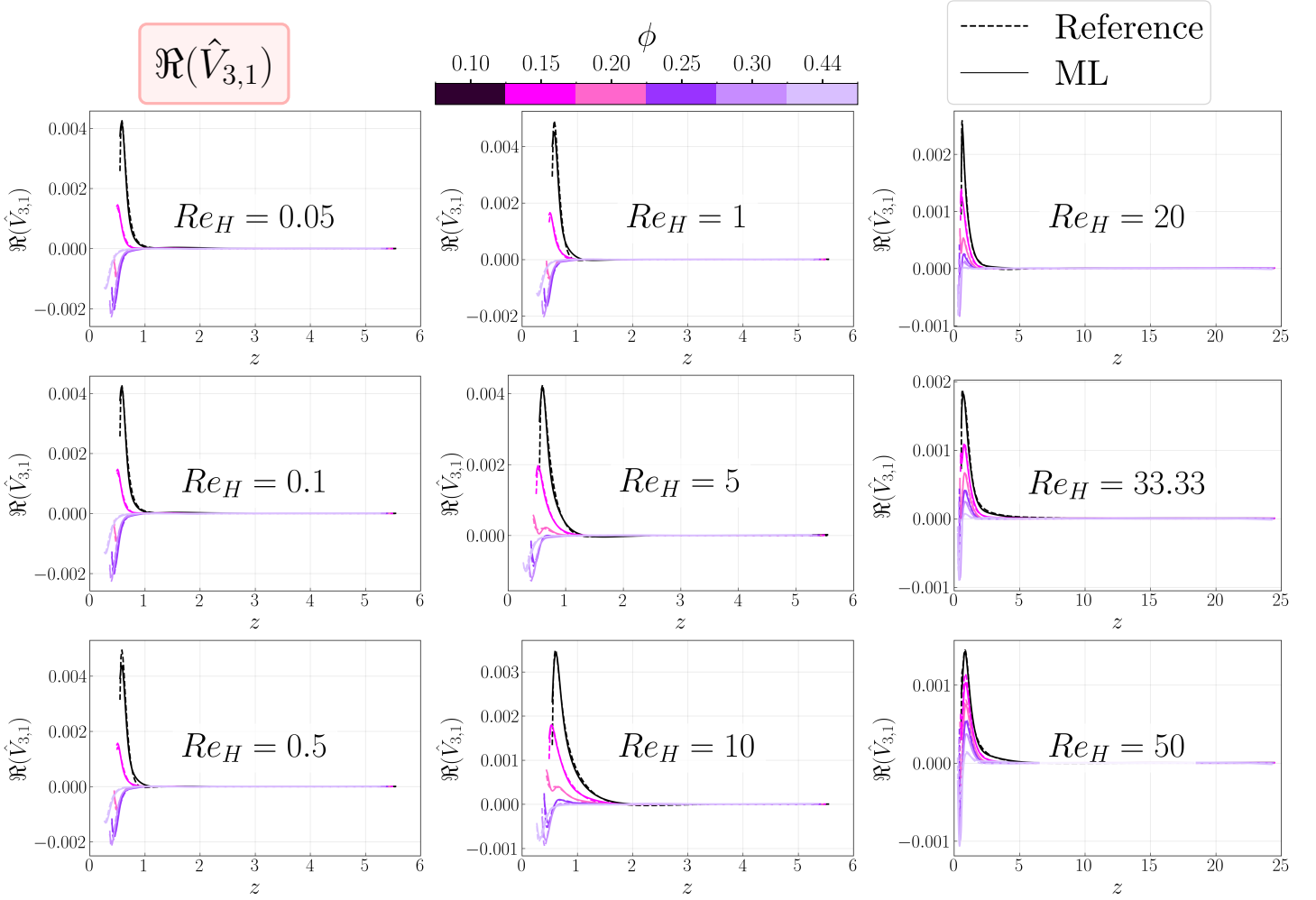


Figure C.38: Comparison of the reference DFT coefficient $\Re(\hat{V}_{3,1})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{3,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

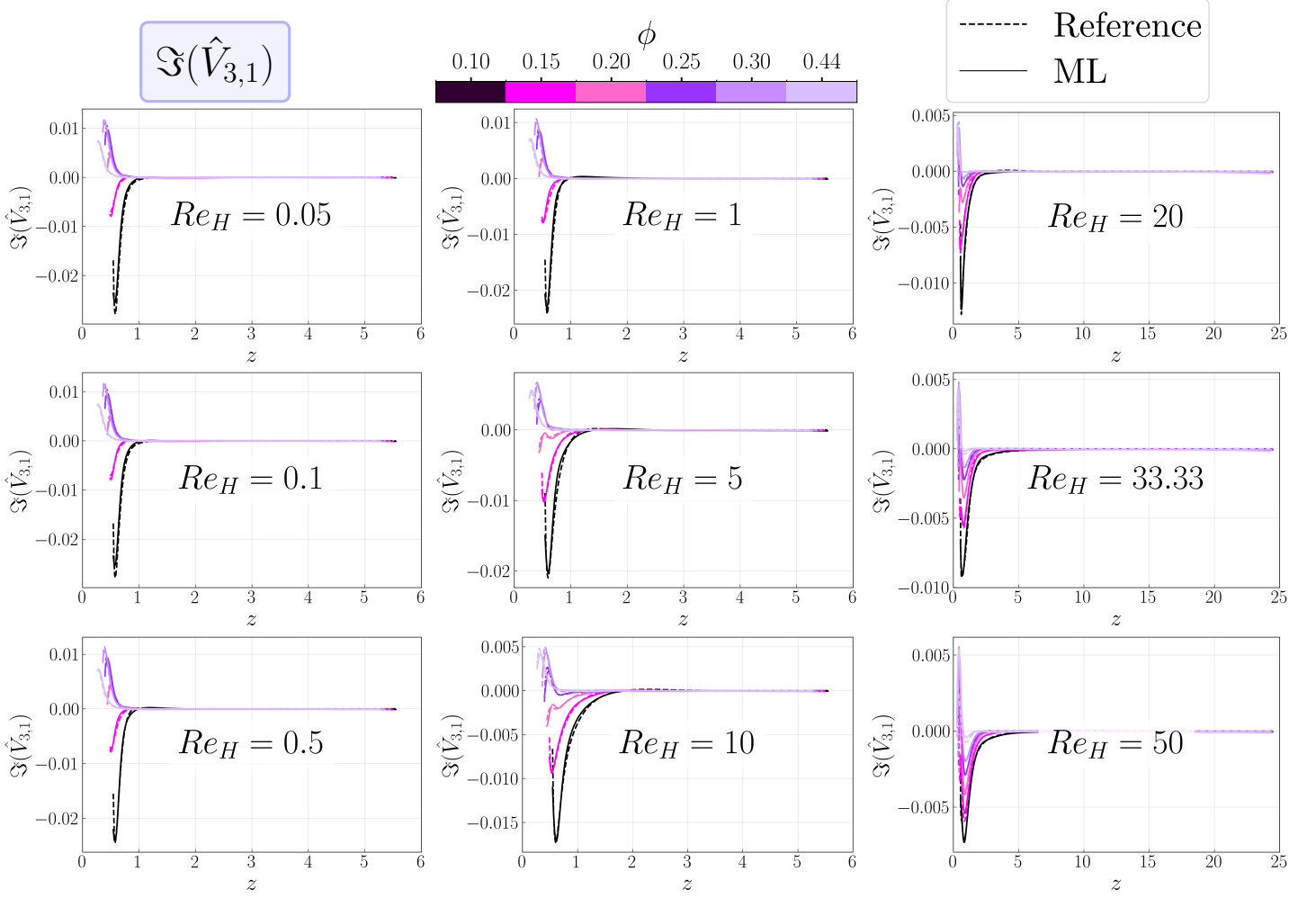


Figure C.39: Comparison of the reference DFT coefficient $\Im(\hat{V}_{3,1})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{3,1}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

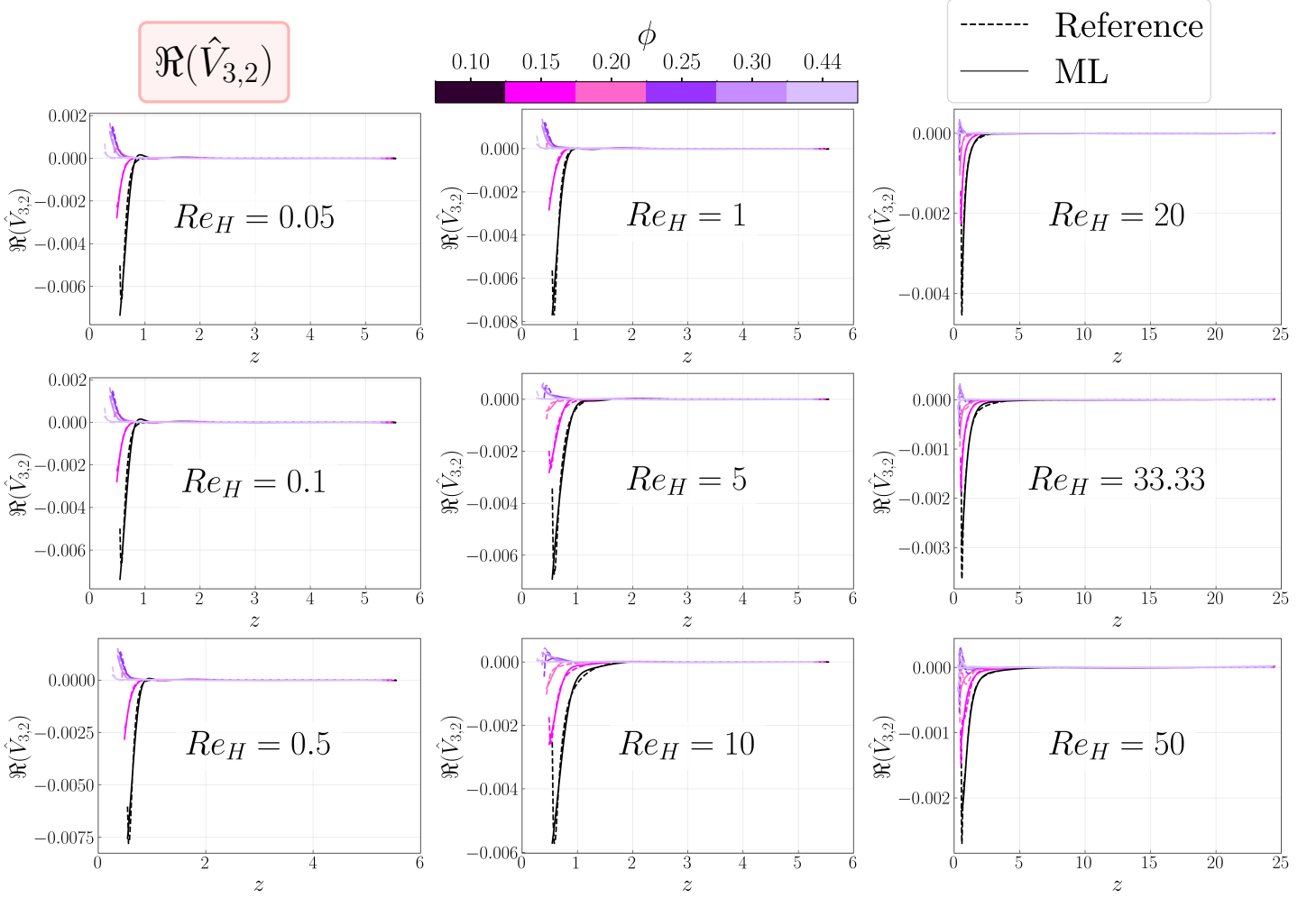


Figure C.40: Comparison of the reference DFT coefficient $\Re(\hat{V}_{3,2})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{3,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

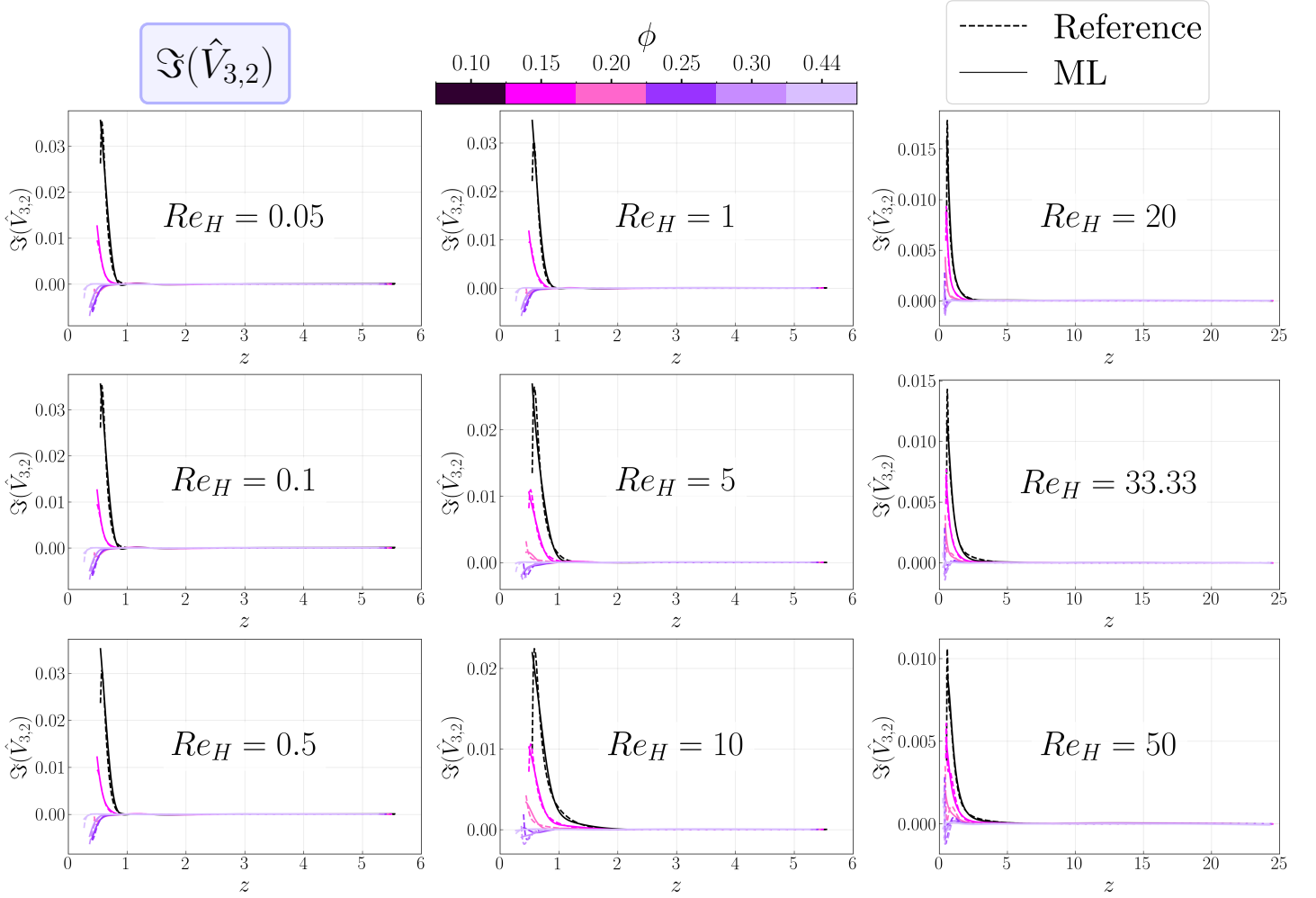


Figure C.41: Comparison of the reference DFT coefficient $\Im(\hat{V}_{3,2})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{3,2}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

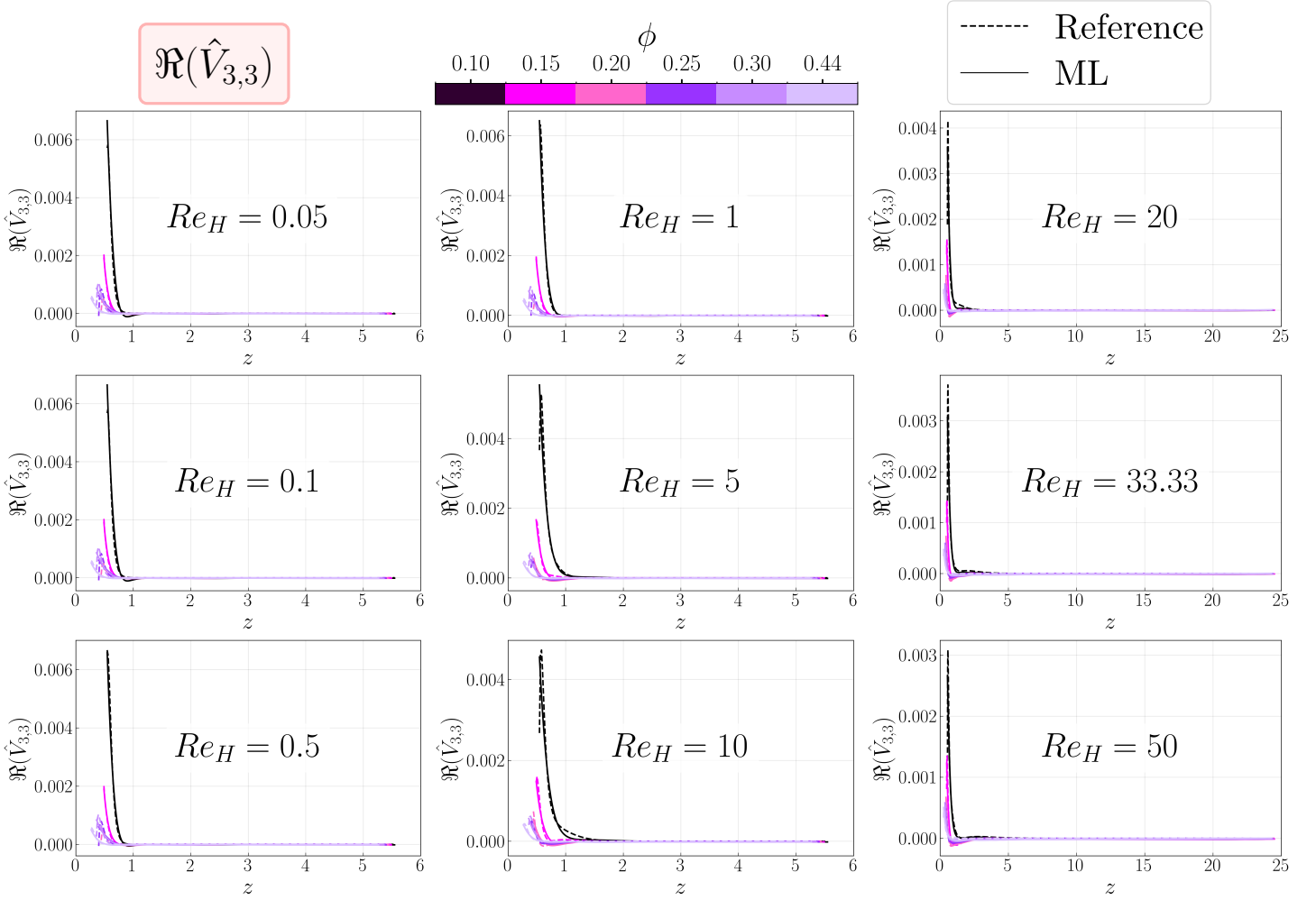


Figure C.42: Comparison of the reference DFT coefficient $\Re(\hat{V}_{3,3})$ (dashed lines) and its neural-network prediction $\Re(\hat{V}_{3,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.

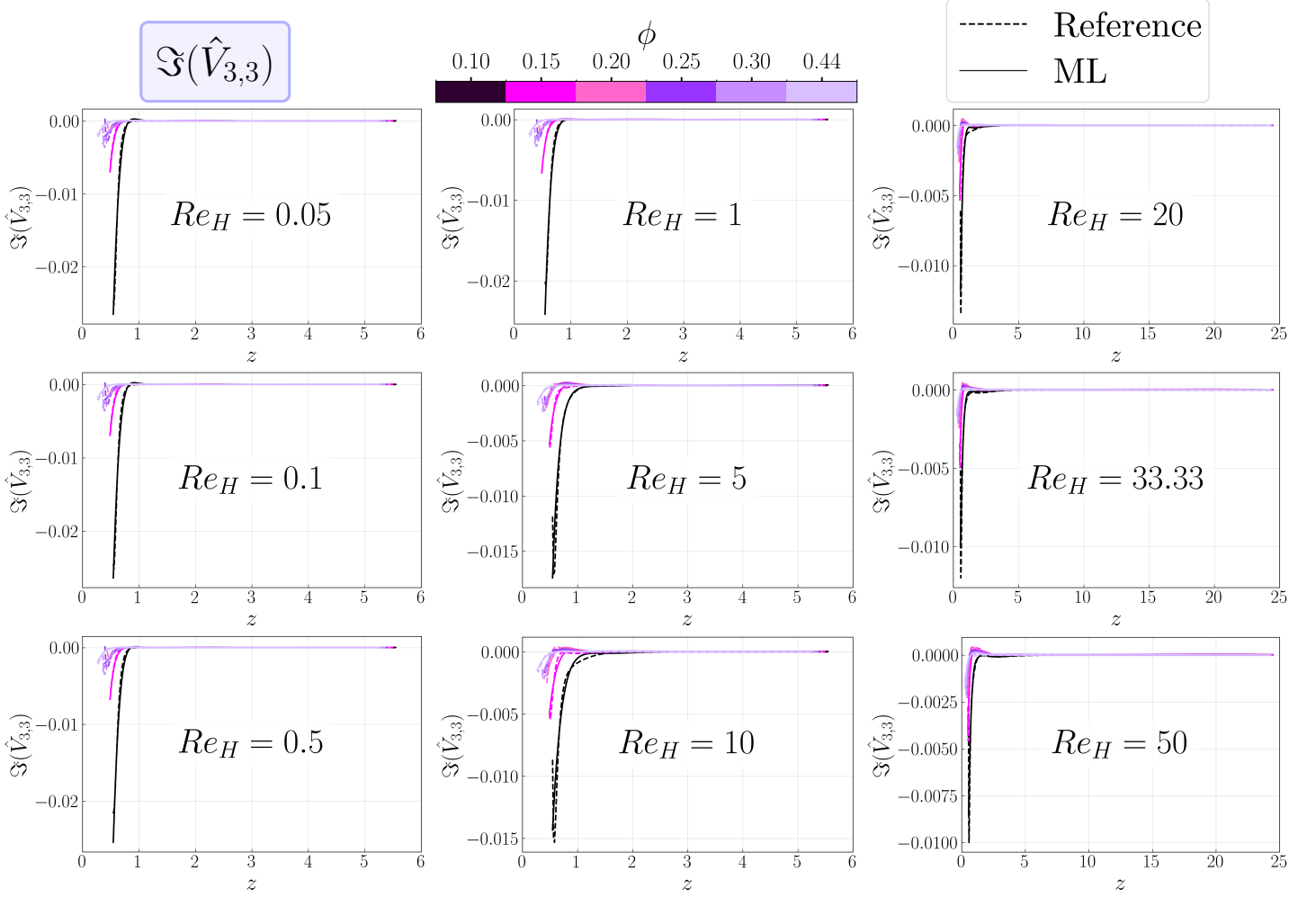


Figure C.43: Comparison of the reference DFT coefficient $\Im(\hat{V}_{3,3})$ (dashed lines) and its neural-network prediction $\Im(\hat{V}_{3,3}^{\text{ML}})$ (solid lines) downstream of the SRG. Each panel shows the coefficients as functions of z at the indicated Reynolds number; colors denote the porosity ϕ . All cases lie within the parameter range covered by the training database.